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Q.1. Given the three integers n, a and b return the nth magical number, since the answer is very large mod $10^9 + 7$.

A positive integer is magical number when the number is divisible by either a or b.

n = 1 a = 2, b = 3
Out = 2

n = 4, a = 2, b = 3
out = 6

→ Optimal :-

Find x such that count of magical numbers <= x

Use Binary search from 1 to n*min(a,b)

```
if (count < n)
    l = mid+1
else
    r = mid
```

$$count = \frac{x}{a} + \frac{x}{b} - \frac{x}{lcm(a,b)}$$

```
Long Long gcd(long long x, long long y){
    return y == 0 ? x : gcd(y, x%y);
}

int nthMagicalNumber(int n, int a, int b){
    const int MOD = 1e9+7;

    long long lcm = (a*b)/gcd(a,b);

    long long low = 1;
    long long high = n* min(a,b);

    while(low< high){
        long long mid = low +(high - low)/2;

        long long count = mid/a + mid/b - mid/lcm;

        if(count <n){
            low = mid+1;
        }else {
            high = mid;
        }
    }
    return low % MOD;
}
```